Material Design 3 & Material 3 Expressive Design System Guide

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Purpose: Comprehensive design system guide for AI-assisted Material Design 3 implementation

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Core Design Principles

Principle 1: Material is the Metaphor

RULE: Design should feel tangible, responsive, and grounded in physical principles.

Core Tenets:

- **Surface-based**: UI built on surfaces with elevation and depth
- **Color with purpose**: Every color choice communicates meaning
- **Motion with meaning**: Animation clarifies relationships and feedback
- **Readable typography**: Text is the primary content and should be accessible

Implementation:

- Use elevation to create depth hierarchy

- Apply color semantically (primary for actions, error for destructive)
- Include subtle motion for state changes
- Ensure typography scales for readability

Principle 2: User-Centric Adaptability

RULE: Design adapts to user needs, preferences, and device context.

What This Means:

- **Dynamic color**: Adapt to user's system color palette
- **Responsive layout**: Scale gracefully across devices
- **Accessibility first**: Color contrast, keyboard navigation, screen readers
- **User preference support**: Dark mode, reduced motion, increased text size

Never:

- Use color alone to communicate critical information
- Rely on motion for essential functionality
- Ignore keyboard navigation
- Force fixed layouts that don't respond

Principle 3: Expression Through Customization

RULE: Material 3 Expressive extends Material 3 with bolder, more distinctive styling.

Expressive vs. Baseline:

- **Baseline Material 3**: Subtle, refined, professional
- **Expressive Material 3**: Bold shapes, vibrant color, personality-driven

When to Use Each:

- **Baseline**: Enterprise, professional, data-heavy applications
- **Expressive**: Creative, lifestyle, branded consumer applications

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Color System & Dynamic Color

Rule 1: Color Roles and Semantics

RULE: Use semantic color roles instead of arbitrary color names.

Primary Color Role:

- Purpose: Primary interactive elements (buttons, links, highlights)
- Use: Call-to-action buttons, selected states, active elements
- Accessibility: Must meet 4.5:1 contrast with white or black

Secondary Color Role:

- Purpose: Supporting actions and less critical elements
- Use: Secondary buttons, chips, dividers
- Note: Less prominent than primary

Tertiary Color Role:

- Purpose: Filter chips, linked elements (new in Material 3)
- Use: Tertiary buttons, highlighted search results
- Accessibility: Alternative path for interactions

Error/Success/Warning Colors:

- **Error**: Destructive actions, form errors, alerts (typically red/pink)
- **Success**: Positive confirmations, validations (typically green)
- **Warning**: Cautionary states, important notices (typically orange/amber)

✅ **CORRECT**:
primary = #6750A4
onPrimary = #FFFFFF
primaryContainer = #EADDFF
onPrimaryContainer = #21005E

secondary = #625B71
onSecondary = #FFFFFF
secondaryContainer = #E8DEF8
onSecondaryContainer = #1E192B

error = #B3261E
onError = #FFFFFF
errorContainer = #F9DEDC
onErrorContainer = #410E0B

❌ **WRONG**:
accentColor = #6750A4 // Vague, not semantic
darkBlue = #003D82 // Arbitrary naming
highlight = #FFE082 // Doesn't indicate purpose

Rule 2: Neutral Colors (Surface & On Surface)

RULE: Use neutral colors for backgrounds, surfaces, and text.

Surface Role:

- Default container for content
- Should have sufficient contrast with text
- Multiple surface tones (surface, surfaceDim, surfaceBright)

On Surface Role:

- Text color on surfaces
- Ensures readability at all sizes
- Adapts in dark mode

✅ **CORRECT**:
surface = #FFFBFE
surfaceDim = #DDD7E0
surfaceBright = #FFFBFE

onSurface = #1C1B1F
onSurfaceVariant = #49454E

Rule 3: Dynamic Color System

RULE: Implement dynamic color that adapts to user's system accent color (Android 12+, iOS 15+).

Implementation Steps:

1. Extract user's system accent color
2. Apply tone-based color scaling
3. Maintain WCAG AA contrast ratios
4. Support both light and dark modes

Dynamic Color Process:

- System provides base hue
- Material generates tonal palette (0-100 scale)
- Select tones for each role maintaining contrast
- No hardcoded colors - all generated from system

✅ **CORRECT**:
```
// Android Compose
val primary = dynamicColorScheme.primary // System-derived
val secondary = dynamicColorScheme.secondary
// Colors adapt to system accent color
```

❌ **WRONG**:
```
// Ignoring dynamic color
val primary = Color(0xFF6750A4) // Static color, ignores system preference
```

Rule 4: Color Contrast Requirements

RULE: Maintain WCAG AA minimum contrast ratios.

Standards:

- **4.5:1**: Normal text (14px and below)
- **3:1**: Large text (18px+, bold 14px+)
- **3:1**: UI components and graphical elements

Verification:

- Use contrast checking tools (WebAIM, Stark, etc.)
- Test in both light and dark modes
- Verify with actual text colors

✅ **CORRECT**:
Primary #6750A4 on White #FFFFFF
Contrast: 5.2:1 ✓ Exceeds 4.5:1

On Surface #1C1B1F on Surface #FFFBFE
Contrast: 16:1 ✓ Maximum contrast

✕ **WRONG**:
Primary #A189BD (too light) on White #FFFFFF
Contrast: 2.1:1 ✗ Below 4.5:1 minimum

Typography System

Rule 1: Type Scale Structure

RULE: Use Material 3's predefined type scale with consistent sizing and weight.

Material 3 Type Scale:

Role	Size	Weight	Line Height	Letter Spacing
Display Large	57px	Regular (400)	64px	0px
Display Medium	45px	Regular (400)	52px	0px
Display Small	36px	Regular (400)	44px	0px
Headline Large	32px	Regular (400)	40px	0px
Headline Medium	28px	Regular (400)	36px	0px
Headline Small	24px	Regular (400)	32px	0px
Title Large	22px	Medium (500)	28px	0px
Title Medium	16px	Medium (500)	24px	0.15px
Title Small	14px	Medium (500)	20px	0.1px
Body Large	16px	Regular (400)	24px	0.15px
Body Medium	14px	Regular (400)	20px	0.25px
Body Small	12px	Regular (400)	16px	0.4px
Label Large	14px	Medium (500)	20px	0.1px
Label Medium	12px	Medium (500)	16px	0.5px
Label Small	11px	Medium (500)	16px	0.5px

✅ **CORRECT**:
```
// Compose
Text(
text = "Headline Large",
style = MaterialTheme.typography.headlineLarge // 32px, Regular
)

Text(
text = "Body Medium",
style = MaterialTheme.typography.bodyMedium // 14px, Regular
)
```

❌ **WRONG**:
```
// Arbitrary sizes and weights
Text(text = "Heading", fontSize = 20.sp, fontWeight = FontWeight.Bold)
Text(text = "Body", fontSize = 13.sp) // Not standard scale
```

Rule 2: Font Family Selection

RULE: Use system fonts or Material font stack for optimal performance.

Material 3 Recommendations:

- **Default**: Roboto (Android), -apple-system (iOS/Web)
- **Expressive**: Pair Roboto with expressive serif for headlines

✅ **CORRECT**:
```
/* Web */
font-family: Roboto, -apple-system, BlinkMacSystemFont, "Segoe UI", sans-serif;

/* Expressive pairing */
h1 { font-family: Georgia, serif; }
body { font-family: Roboto, sans-serif; }
```

❌ **WRONG**:
```
/* Heavy custom fonts slow down performance */
font-family: "CustomExpensiveFont", serif;
```

Rule 3: Line Height and Letter Spacing

RULE: Follow Material 3 specifications for line height and letter spacing.

Guidelines:

- **Display sizes**: 1.12x line height ratio (57px × 1.12 = ~64px)
- **Headline sizes**: 1.25x ratio
- **Title sizes**: 1.27x ratio
- **Body text**: 1.5x ratio (standard for readability)
- **Letter spacing**: Increase for smaller sizes (Label Small: 0.5px)

✅ **CORRECT**:
```
/* Body text - standard readability /
p {
```

```
font-size: 16px;
line-height: 24px; / 1.5x ratio */
letter-spacing: 0.15px;
}

/* Display text - tighter /
h1 {
font-size: 57px;
line-height: 64px; / 1.12x ratio */
letter-spacing: 0px;
}
```

Rule 4: Text Truncation and Overflow

RULE: Handle text overflow consistently with Material 3 principles.

Rules:

- Single-line text: Ellipsis (...) at end
- Multi-line text: Fade or truncate
- Body text: Generally no truncation, allow reflow
- Labels: Truncate if necessary

✅ **CORRECT**:

```
/* Single line with ellipsis */
.single-line {
white-space: nowrap;
overflow: hidden;
text-overflow: ellipsis;
}

/* Multi-line fade */
.multi-line {
display: -webkit-box;
-webkit-line-clamp: 2;
-webkit-box-orient: vertical;
overflow: hidden;
}
```

Spacing & Layout Grid

Rule 1: 4px Base Grid System

RULE: All spacing uses 4px as the base unit.

Spacing Scale:

- **4px**: Minimal spacing (tight pairs)
- **8px**: Small spacing (common)
- **12px**: Small-medium spacing
- **16px**: Medium spacing (very common)
- **20px**: Medium-large spacing

- **24px**: Large spacing (common)
- **32px**: Extra large spacing
- **48px**: Extra extra large spacing
- **56px**: Maximum (rare)

Multiples of 4: All spacing = 4n (4, 8, 12, 16, 20, 24, 28, 32...)

✅ **CORRECT**:
/* Valid spacing */
padding: 16px; /* 4 × 4 */
margin-bottom: 24px; /* 4 × 6 */
gap: 8px; /* 4 × 2 */

/* Button with proper spacing */
button {
padding: 12px 24px; /* Vertical: 3×4, Horizontal: 6×4 */
gap: 8px; /* Between icon and text */
}

❌ **WRONG**:
/* Non-4px multiples */
padding: 10px; /* Not divisible by 4 */
margin: 15px; /* Not divisible by 4 */
gap: 7px; /* Not divisible by 4 */

Rule 2: Container and Component Spacing

RULE: Containers have consistent padding, components have consistent internal spacing.

Container Padding (by screen size):

- **Mobile**: 16px horizontal
- **Tablet**: 24px horizontal
- **Desktop**: 24px horizontal (or more for ultra-wide)

Component Spacing:

- **Button**: 12px vertical, 24px horizontal (icon buttons: 12px)
- **Card**: 24px padding
- **Dialog**: 24px padding
- **Text field**: 16px horizontal, 12px top/bottom

✅ **CORRECT**:
/* Container */
.screen-content {
padding: 0 16px; /* Mobile standard */
}

/* Card component */
.card {
padding: 24px;
margin-bottom: 16px;
}

/* Text field /
.text-field {
padding: 12px 16px;
height: 56px; / Standard touch target */*
}

Rule 3: Responsive Layout Breakpoints

RULE: Use Material 3 standard breakpoints for responsive design.

Breakpoint System:

- **Compact**: < 600dp (phones)
- **Medium**: 600dp - 839dp (tablets, small)
- **Expanded**: ≥ 840dp (tablets large, desktops)

Layout Columns:

- **Compact**: 4 columns
- **Medium**: 8 columns
- **Expanded**: 12 columns

✅ **CORRECT**:
/* Compact (mobile) - 4 column grid */
@media (max-width: 599px) {
.grid {
grid-template-columns: repeat(4, 1fr);
padding: 0 16px;
}
}

/* Medium (tablet) - 8 column grid */
@media (min-width: 600px) and (max-width: 839px) {
.grid {
grid-template-columns: repeat(8, 1fr);
padding: 0 24px;
}
}

/* Expanded (desktop) - 12 column grid */
@media (min-width: 840px) {
.grid {
grid-template-columns: repeat(12, 1fr);
padding: 0 24px;
}
}

Rule 4: Gutters and Column Gaps

RULE: Use consistent gutter width between columns and rows.

Gutter Width:

- **Standard**: 16px between columns
- **Compact**: 8px (rare, only when space critical)
- **Row gap**: Typically 16px or 24px

✅ **CORRECT**:
```
.grid {
display: grid;
grid-template-columns: repeat(auto-fit, minmax(300px, 1fr));
gap: 16px; /* Both column and row gap */
}

/* Or explicit separate gaps */
.grid {
column-gap: 16px;
row-gap: 24px;
}
```

Shape & Elevation

Rule 1: Corner Radius System

RULE: Use Material 3's corner radius scale (0px, 4px, 8px, 12px, 16px, 28px, 32px).

Shape Tokens:

- **None (0px)**: Sharp, no border radius
- **Extra Small (4px)**: Small components, slight softness
- **Small (8px)**: Standard components
- **Medium (12px)**: Cards, elevated components
- **Large (16px)**: Large containers
- **Extra Large (28px)**: Prominent, distinctive elements
- **Full (32px)**: Fully rounded shapes

Component Usage:

- **Buttons**: 8px or 28px (Material 3 uses 8px standard)
- **Cards**: 12px
- **Dialogs**: 28px
- **FAB**: Fully rounded (circle)
- **Text fields**: 8px or 12px
- **Chips**: 8px

✅ **CORRECT**:
```
/* Standard button */
button {
border-radius: 8px;
}
```

```
/* Card */
.card {
border-radius: 12px;
}

/* Dialog - more prominent */
.dialog {
border-radius: 28px;
}

/* FAB - fully rounded /
.fab {
border-radius: 50%; / Full circle */
width: 56px;
height: 56px;
}
```

✕ **WRONG**:

```
/* Non-standard radii /
button { border-radius: 6px; } / Not Material 3 scale /
card { border-radius: 15px; } / Not Material 3 scale */
```

Rule 2: Elevation System (Material 3)

RULE: Use elevation to create depth and hierarchy using shadows and scale.

Elevation Levels:

- **Level 0**: No shadow (flush)
- **Level 1**: Subtle shadow (small elevation)
- **Level 2**: Noticeable shadow (medium elevation)
- **Level 3**: Clear shadow (prominent)
- **Level 4**: Strong shadow (floating)
- **Level 5**: Pronounced shadow (dialog, highest)

Shadow Implementation:

```
/* Level 0 */
box-shadow: none;

/* Level 1 */
box-shadow: 0px 1px 3px rgba(0, 0, 0, 0.12), 0px 1px 2px rgba(0, 0, 0, 0.24);

/* Level 2 */
box-shadow: 0px 3px 6px rgba(0, 0, 0, 0.15), 0px 2px 4px rgba(0, 0, 0, 0.12);

/* Level 3 */
box-shadow: 0px 10px 20px rgba(0, 0, 0, 0.15), 0px 3px 6px rgba(0, 0, 0, 0.10);

/* Level 4 */
box-shadow: 0px 15px 30px rgba(0, 0, 0, 0.20), 0px 5px 10px rgba(0, 0, 0, 0.12);

/* Level 5 */
box-shadow: 0px 20px 40px rgba(0, 0, 0, 0.25), 0px 8px 16px rgba(0, 0, 0, 0.15);
```

Component Elevation:

- **Flat**: Buttons, chips, text fields (level 0-1)
- **Raised**: Card, navigation bar (level 1-2)
- **Floating**: FAB, bottom sheet, modal (level 3-4)
- **Modal**: Dialog (level 5)

✅ **CORRECT**:
```
/* Card - elevation 1 */
.card {
border-radius: 12px;
box-shadow: 0px 1px 3px rgba(0, 0, 0, 0.12), 0px 1px 2px rgba(0, 0, 0, 0.24);
}

/* FAB - elevation 4 */
.fab {
box-shadow: 0px 15px 30px rgba(0, 0, 0, 0.20), 0px 5px 10px rgba(0, 0, 0, 0.12);
}

/* Dialog - elevation 5 */
.dialog {
box-shadow: 0px 20px 40px rgba(0, 0, 0, 0.25), 0px 8px 16px rgba(0, 0, 0, 0.15);
}
```

Motion & Transitions

Rule 1: Duration Guidelines

RULE: Use appropriate animation durations for different interaction types.

Duration Scale:

- **Short (50ms)**: Micro-interactions (ripple, hover)
- **Medium (200ms)**: Standard transitions (state changes)
- **Long (300ms)**: Navigation, complex animations
- **Extra Long (500ms+)**: Rare, only for special cases

Formula: Longer animations = more complex visual change

✅ **CORRECT**:
```
/* Hover state - short */
button:hover {
transition: background-color 200ms cubic-bezier(0.4, 0, 0.2, 1);
}

/* Page transition - medium-long */
.page-exit {
animation: slideOut 300ms cubic-bezier(0.4, 0, 0.2, 1);
}

/* Loading animation - longer loop */
@keyframes spin {
0% { transform: rotate(0deg); }
```

```
100% { transform: rotate(360deg); }
}
.spinner { animation: spin 1.2s linear infinite; }
```

Rule 2: Easing Functions

RULE: Use Material 3 standard easing for consistent motion feel.

Standard Easing: `cubic-bezier(0.4, 0, 0.2, 1)`

- Used for most animations
- Natural, responsive feel
- Works for all transition types

Decelerate Easing: `cubic-bezier(0, 0, 0.2, 1)`

- Used when element enters
- Slows as it settles
- Emphasizes arrival

Accelerate Easing: `cubic-bezier(0.4, 0, 1, 1)`

- Used when element exits
- Accelerates departure
- Emphasizes movement away

✅ **CORRECT**:

```
/* Standard - most common */
transition: all 200ms cubic-bezier(0.4, 0, 0.2, 1);

/* Element entering */
.fade-in {
animation: fadeIn 300ms cubic-bezier(0, 0, 0.2, 1);
}

/* Element exiting */
.fade-out {
animation: fadeOut 300ms cubic-bezier(0.4, 0, 1, 1);
}

@keyframes fadeIn {
from { opacity: 0; }
to { opacity: 1; }
}
```

Rule 3: Motion Principles

RULE: Motion must serve a purpose - clarify, guide, or delight.

No Gratuitous Animation:

- Every animation should communicate something
- Avoid animation for decoration alone

- Consider `prefers-reduced-motion`

✅ **CORRECT**:
```
/* Meaningful: Shows state change */
button {
transition: background-color 200ms;
}
button:active {
background-color: var(--color-primary-active);
}

/* Meaningful: Guides user attention */
@keyframes pulse {
0%, 100% { opacity: 1; }
50% { opacity: 0.5; }
}
.notification {
animation: pulse 2s infinite;
}

/* Respect user preferences */
@media (prefers-reduced-motion: reduce) {
* {
animation-duration: 0.01ms !important;
animation-iteration-count: 1 !important;
transition-duration: 0.01ms !important;
}
}
```

Components Overview

Rule 1: Button Variants

RULE: Material 3 defines four button styles, each with specific use cases.

Filled Button:

- **Primary action** in a screen
- Highest emphasis
- Use sparingly (typically one per screen)
- Default button style

```
.button-filled {
background-color: var(--md-sys-color-primary);
color: var(--md-sys-color-on-primary);
padding: 12px 24px;
border: none;
border-radius: 8px;
}

.button-filled:hover {
background-color: var(--md-sys-color-primary-hover);
```

```
}

.button-filled:active {
background-color: var(--md-sys-color-primary-active);
}
```

Filled Tonal Button:

- **Secondary action**
- Medium emphasis
- Useful for multiple actions on a screen

```
.button-tonal {
background-color: var(--md-sys-color-secondary-container);
color: var(--md-sys-color-on-secondary-container);
padding: 12px 24px;
border: none;
border-radius: 8px;
}
```

Outlined Button:

- **Lower priority** actions
- Lower emphasis
- Often paired with filled buttons

```
.button-outlined {
background-color: transparent;
color: var(--md-sys-color-primary);
padding: 12px 24px;
border: 1px solid var(--md-sys-color-outline);
border-radius: 8px;
}
```

Text Button:

- **Lowest emphasis** actions
- Minimal visual weight
- Use in constrained spaces

```
.button-text {
background-color: transparent;
color: var(--md-sys-color-primary);
padding: 12px 8px;
border: none;
text-decoration: underline;
}
```

Rule 2: Text Field / Input Component

RULE: Material 3 text fields have consistent sizing and styling.

Standard Text Field:

- **Height**: 56px (including padding)
- **Label**: Floats above on focus/fill
- **Supporting text**: Below field
- **Trailing icon**: Clear button, optional

```
.text-field {
display: flex;
flex-direction: column;
gap: 4px;
width: 100%;
max-width: 360px;
}

.text-field-input {
height: 56px;
padding: 12px 16px;
border: 1px solid var(--md-sys-color-outline);
border-radius: 8px;
font-size: 16px;
font-family: var(--md-font-family);
}

.text-field-input:focus {
border-color: var(--md-sys-color-primary);
outline: none;
}

.text-field-label {
font-size: 12px;
color: var(--md-sys-color-on-surface-variant);
font-weight: 500;
}

.text-field-supporting-text {
font-size: 12px;
color: var(--md-sys-color-on-surface-variant);
margin-top: 4px;
}
```

Rule 3: Card Component

RULE: Cards are containers for related content with consistent styling.

Filled Card (default):

- Background color = primary surface
- Elevation = 0 (flat)

- Use for content collection

```
.card-filled {
background-color: var(--md-sys-color-surface);
border-radius: 12px;
padding: 24px;
border: 1px solid var(--md-sys-color-outline);
}
```

Elevated Card:

- Has elevation shadow
- Use to emphasize certain cards

```
.card-elevated {
background-color: var(--md-sys-color-surface);
border-radius: 12px;
padding: 24px;
box-shadow: 0px 1px 3px rgba(0, 0, 0, 0.12), 0px 1px 2px rgba(0, 0, 0, 0.24);
}
```

Outlined Card:

- Has border instead of fill or shadow
- Lower emphasis
- Use for alternative styling

```
.card-outlined {
background-color: transparent;
border: 1px solid var(--md-sys-color-outline);
border-radius: 12px;
padding: 24px;
}
```

Component States

Rule 1: Interactive States

RULE: Each interactive component must support these states visually and accessibly.

States:

- **Enabled**: Default, interactive state
- **Hovered**: Subtle background change
- **Focused**: Focus ring visible (keyboard nav)
- **Pressed/Active**: Darker or inverted color
- **Disabled**: Reduced opacity (38%), not interactive

✅ **CORRECT**:

```
button {
opacity: 1;
transition: all 200ms cubic-bezier(0.4, 0, 0.2, 1);
}
```

```
button:hover {
background-color: rgba(var(--color-primary-rgb), 0.08); /* 8% overlay */
}

button:focus {
outline: 2px solid var(--color-primary);
outline-offset: 2px;
}

button:active {
background-color: rgba(var(--color-primary-rgb), 0.12); /* 12% overlay */
}

button:disabled {
opacity: 0.38;
cursor: not-allowed;
}
```

Rule 2: Form Validation States

RULE: Text fields and form components must clearly indicate validation state.

Valid State:

- No error message
- Optional: Green checkmark
- Border = default outline color

Invalid State:

- Border color = error color
- Error message displayed below field
- Supporting text shows error details

Warning State:

- Border color = warning color
- Supporting text shows warning details
- Allows submission (user decision)

✅ **CORRECT**:

```
/* Valid */
.text-field.valid {
border-color: var(--md-sys-color-outline);
}

/* Invalid */
.text-field.invalid {
border-color: var(--md-sys-color-error);
}

.text-field.invalid ~ .error-text {
color: var(--md-sys-color-error);
```

```
display: block;
}

/* Warning */
.text-field.warning {
border-color: var(--md-sys-color-warning);
}
```

Accessibility Standards

Rule 1: WCAG 2.1 Level AA Compliance

RULE: All interfaces must meet WCAG 2.1 Level AA minimum.

Color Contrast:

- 4.5:1 for normal text
- 3:1 for large text (18px+)
- 3:1 for UI components

Touch Target Size:

- Minimum 44×44 dp (44 × 44 pixels)
- Mobile: 48×48 dp recommended
- Desktop: 32×32 dp acceptable

Focus Management:

- Visible focus indicator (focus ring)
- Logical tab order
- Skip navigation links

✅ **CORRECT**:

```
/* Focus indicator */
:focus-visible {
outline: 2px solid var(--color-primary);
outline-offset: 2px;
}

/* Touch target */
button {
min-width: 44px;
min-height: 44px;
}

/* Color not sole differentiator /
input.error {
border: 2px solid var(--color-error);
/ Also text message, not color alone */
}
```

Rule 2: Semantic HTML and ARIA

RULE: Use semantic HTML and ARIA attributes for screen reader support.

Semantic Elements:

- `<button>` for buttons (not `<div>`)
- `<a>` for links
- `<form>` for forms
- `<label>` for form labels
- `<nav>` for navigation

ARIA Attributes:

- `aria-label`: For icon-only buttons
- `aria-expanded`: For collapsible sections
- `aria-hidden="true"`: For decorative elements
- `role`: When semantic HTML unavailable

✅ **CORRECT**:

× Link text

Email:
Custom button

✗ **WRONG**:

Click me
Button

Rule 3: Keyboard Navigation

RULE: All functionality accessible via keyboard.

Requirements:

- Tab order logical and visible
- Enter/Space activate buttons
- Arrow keys for selection
- Escape closes modals
- No keyboard traps

✅ **CORRECT**:

Save Cancel

Content

Material 3 Expressive Extensions

Rule 1: When to Use Expressive

RULE: Use Material 3 Expressive for applications that benefit from bold, distinctive design.

Good Candidates for Expressive:

- Creative/design-focused apps
- Consumer lifestyle apps
- Branded applications
- Entertainment apps
- Young audience focus

Stick with Baseline for:

- Enterprise applications
- Data-heavy interfaces
- Professional tools
- Accessibility-critical apps

Rule 2: Expressive Shape System

RULE: Expressive uses bolder corner radius for distinctive appearance.

Expressive Shape Scale (vs. Baseline):

- **None (0px)**: Same
- **Extra Small (8px → 12px)**: Slightly bolder
- **Small (12px → 16px)**: Notably rounder
- **Medium (16px → 24px)**: Much rounder
- **Large (24px → 32px)**: Very pronounced
- **Extra Large (32px → 48px)**: Bold, sculptural
- **Full (48px+)**: Organic, flowing

✅ **CORRECT EXPRESSIVE**:
/* Expressive shapes - bolder /
.button-expressive {
border-radius: 12px; / vs 8px in baseline */
}

.card-expressive {
border-radius: 24px; /* vs 12px in baseline */
}

.dialog-expressive {
border-radius: 32px; /* vs 28px in baseline */
}

Rule 3: Expressive Color Application

RULE: Expressive encourages bolder, more saturated color usage.

Expressive Color Strategies:

1. **Increase saturation**: Use more vibrant hues
2. **Surface variation**: Distinct surface colors for sections
3. **Color layering**: Multiple color uses per screen
4. **Custom palette**: Brand colors integrated more prominently

✅ **CORRECT EXPRESSIVE**:
/* Baseline - restrained */
:root {
--md-sys-color-primary: #6750A4;
}

/* Expressive - bolder /
:root {
--md-sys-color-primary: #5A4C9E; / Slightly deeper /
--md-sys-color-secondary: #795548; / More saturated /
--md-sys-color-surface-1: #F3E5F5; / Colored surface */
}

Rule 4: Expressive Typography Enhancement

RULE: Expressive typography is more distinctive and personality-driven.

Expressive Typography Changes:

- Headlines use serif fonts or distinctive weights
- Body text may use custom font pairings
- Letter spacing more generous in headlines
- Weight contrast more pronounced

✅ **CORRECT EXPRESSIVE**:
/* Baseline - neutral */
h1 { font-family: Roboto, sans-serif; }

/* Expressive - personality */
h1 {
font-family: "Georgia", serif;
font-weight: 700;
letter-spacing: 0.5px;
}

/* Display text - more expressive */
.display-large {
font-family: "Georgia", serif;
font-size: 72px;
line-height: 80px;
}

Dark Mode Implementation

Rule 1: Color Mapping in Dark Mode

RULE: Dark mode colors are specifically designed to reduce eye strain and maintain readability.

Dark Mode Color Roles:

- **Background**: Darker surface (typically #121212)
- **Surface**: Slightly lighter than background
- **Primary**: Lighter version of light mode primary
- **On Surface**: Lighter text (typically #E1E0E6)
- **Outline**: Lighter than light mode
- **Error**: Lighter red/pink (better contrast on dark)

✅ **CORRECT**:
```
@media (prefers-color-scheme: dark) {
:root {
--md-sys-color-background: #121212;
--md-sys-color-surface: #1C1B1F;
--md-sys-color-primary: #B39DDB; /* Lighter than light mode */
--md-sys-color-on-surface: #E1E0E6;
--md-sys-color-outline: #49454F;
--md-sys-color-error: #F2B8B5;
}
}
```

Rule 2: Implementation Methods

RULE: Support dark mode through CSS media query and data attribute.

Method 1: System Preference:
```
@media (prefers-color-scheme: dark) {
:root { /* dark colors */ }
}
```

Method 2: User Toggle (data attribute):

```
<html data-color-scheme="dark">
```

```
html[data-color-scheme="dark"] {
/* dark colors */
}
```

Method 3: JavaScript Detection:
```
const prefersDark = window.matchMedia('(prefers-color-scheme: dark)').matches;
document.documentElement.setAttribute('data-color-scheme',
prefersDark ? 'dark' : 'light'
);
```

Rule 3: Testing Dark Mode

RULE: Test all components thoroughly in both light and dark modes.

Testing Checklist:

- [ ] All text readable in both modes
- [ ] Images not inverted (use filter: invert(0) if needed)
- [ ] SVG icons visible in both modes
- [ ] Contrast ratios maintained (4.5:1 minimum)
- [ ] Focus indicators visible
- [ ] No "flashing" when switching modes

Android Implementation (Compose)

Rule 1: Compose Material3 Theme Setup

RULE: Initialize Material 3 theme with color scheme configuration.

✅ **CORRECT**:
@Composable
fun AppTheme(
useDarkTheme: Boolean = isSystemInDarkTheme(),
content: @Composable () -> Unit
) {
val colorScheme = when {
useDarkTheme -> darkColorScheme()
else -> lightColorScheme()
}

```
MaterialTheme(
    colorScheme = colorScheme,
    typography = Typography(),
    shapes = Shapes(),
    content = content
)
```

}

// In your main activity
@Composable
fun MyApp() {
AppTheme {
MainScreen()
}
}

Rule 2: Using Color Tokens

RULE: Always use MaterialTheme color tokens instead of hardcoded colors.

✅ **CORRECT**:
```
@Composable
fun MyButton() {
Button(
onClick = { },
colors = ButtonDefaults.filledTonalButtonColors()
) {
Text("Click me")
}
}

@Composable
fun MyCard() {
Card(
modifier = Modifier
.fillMaxWidth()
.padding(16.dp),
shape = RoundedCornerShape(12.dp),
colors = CardDefaults.cardColors(
containerColor = MaterialTheme.colorScheme.surface
)
) {
Text(
text = "Card content",
color = MaterialTheme.colorScheme.onSurface,
style = MaterialTheme.typography.bodyMedium
)
}
}
```

✕ **WRONG**:
```
@Composable
fun MyButton() {
Button(
onClick = { },
modifier = Modifier.background(Color(0xFF6750A4)) // Hardcoded
) {
Text("Click me", color = Color.White) // Hardcoded
}
}
```

Rule 3: Spacing and Padding Consistency

RULE: Use Material Design space tokens consistently.

@Composable
fun MyLayout() {
Column(
modifier = Modifier
.fillMaxWidth()
.padding(16.dp) // Standard padding
) {
Text("Title", style = MaterialTheme.typography.headlineSmall)

```
    Spacer(modifier = Modifier.height(8.dp))  // 8dp spacing

    Text("Subtitle", style = MaterialTheme.typography.bodyMedium)

    Spacer(modifier = Modifier.height(16.dp))  // 16dp spacing

    Button(
        onClick = { },
        modifier = Modifier
            .fillMaxWidth()
            .height(56.dp)  // Standard touch target
    ) {
        Text("Action")
    }
}
```

}

Web Implementation (Material Web)

Rule 1: HTML Structure for Material Components

RULE: Use proper semantic HTML with Material Web components.

✅ **CORRECT**:
Filled Button Outlined Button Text Button

Card Title

Card content here

Rule 2: CSS Custom Properties for Theming

RULE: Define and use CSS custom properties for all theme values.

:root {
/* Color tokens */
--md-sys-color-primary: #6750A4;
--md-sys-color-on-primary: #FFFFFF;
--md-sys-color-primary-container: #EADDFF;
--md-sys-color-on-primary-container: #21005E;

```
  --md-sys-color-secondary: #625B71;
  --md-sys-color-on-secondary: #FFFFFF;

  --md-sys-color-surface: #FFFBFE;
  --md-sys-color-on-surface: #1C1B1F;

  /* Typography tokens */
  --md-font-family: "Roboto", -apple-system, sans-serif;
  --md-font-size-body-medium: 14px;
  --md-font-weight-medium: 500;

  /* Spacing tokens */
  --md-space-4: 4px;
  --md-space-8: 8px;
  --md-space-16: 16px;
  --md-space-24: 24px;
```

}

/* Apply tokens */
button {
background-color: var(--md-sys-color-primary);
color: var(--md-sys-color-on-primary);
padding: var(--md-space-12) var(--md-space-24);
font-family: var(--md-font-family);
font-size: var(--md-font-size-body-medium);
}

Theming & Customization

Rule 1: Baseline vs. Custom Color Palettes

RULE: Start with Material Design 3 baseline, customize only when needed for branding.

Customization Process:

1. Use Material Theme Builder (m3.material.io)
2. Select brand color
3. Generate palette for all roles
4. Apply to your project
5. Test contrast and accessibility

✅ **CORRECT**:
/* Using generated palette from Material Theme Builder /
:root {
/ Generated from primary color #6750A4 /
--md-sys-color-primary: #6750A4;
--md-sys-color-primary-container: #EADDFF;
--md-sys-color-on-primary: #FFFFFF;
--md-sys-color-on-primary-container: #21005E;
/ ... all other colors in palette ... */
}

Rule 2: Component Customization

RULE: Customize components conservatively, maintain Material 3 principles.

✅ **CORRECT**:
```
// Compose - customize button
@Composable
fun CustomButton(
onClick: () -> Unit,
text: String,
modifier: Modifier = Modifier
) {
Button(
onClick = onClick,
modifier = modifier,
colors = ButtonDefaults.filledButtonColors(
containerColor = MaterialTheme.colorScheme.primary
),
shape = RoundedCornerShape(8.dp),
elevation = ButtonDefaults.elevation(defaultElevation = 4.dp)
) {
Text(text, style = MaterialTheme.typography.labelLarge)
}
}
```

✕ **WRONG**:
// Completely overriding Material 3

```
Button(
onClick = onClick,
modifier = Modifier
.background(Color.Cyan)
.border(BorderStroke(2.dp, Color.Magenta))
) {
Text("Broken Button", color = Color.Red)
}
```

Design Token System

Rule 1: Token Categories

RULE: Organize tokens into logical categories for maintainability.

Color Tokens:

- Semantic (primary, secondary, error)
- Surface (background, surface, surface variants)
- Component-specific (button, card colors)

Typography Tokens:

- Font families
- Font sizes
- Font weights
- Line heights
- Letter spacing

Spacing Tokens:

- Small (4px, 8px)
- Medium (16px, 24px)
- Large (32px, 48px)

Shape Tokens:

- Corner radius scales (4px through 32px)

Motion Tokens:

- Durations (200ms, 300ms, 500ms)
- Easing functions

Rule 2: Token Naming Convention

RULE: Use consistent, hierarchical naming for tokens.

Format: `--md-[category]-[subcategory]-[state]`

✅ **CORRECT**:
```
/* Color tokens */
--md-sys-color-primary
```

--md-sys-color-on-primary
--md-sys-color-primary-container
--md-sys-color-on-primary-container

/* Typography tokens */
--md-sys-typescale-body-medium-font-family
--md-sys-typescale-body-medium-font-size
--md-sys-typescale-body-medium-font-weight

/* Spacing tokens /
--*md-sys-space-1*
--*md-sys-space-2*
--*md-sys-space-3* / 4px × 3 = 12px */

/* Component tokens (optional) */
--md-comp-button-filled-container-color
--md-comp-button-filled-label-text-color

Best Practices

Best Practice 1: Consistent Component APIs

RULE: All custom components follow Material 3 patterns and prop interfaces.

Standard Props:
// Compose
@Composable
fun MyComponent(
text: String,
onClick: () -> Unit,
enabled: Boolean = true,
modifier: Modifier = Modifier
) { }

// Note: modifier last, enabled is standard state parameter

Best Practice 2: Accessibility Integration

RULE: Never skip accessibility - build it in from the start.

From Day 1:

- Use semantic HTML
- Test keyboard navigation
- Verify color contrast
- Include alt text for images
- Test with screen readers

Best Practice 3: Performance Considerations

RULE: Design system should not compromise app performance.

Optimization:

- Use CSS variables (not computed on every render)
- Lazy-load Material components
- Minimize shadow rendering (expensive)
- Optimize animations for 60fps

Common Mistakes

✖ MISTAKE 1: Using Color Alone for Information

```
/* Wrong - only color indicates error */
input { border-color: red; }

/* Correct - color + icon + text */
.text-field.error {
border-color: var(--md-sys-color-error);
}

.error-icon::before {
content: "⚠ ";
}

.error-text {
color: var(--md-sys-color-error);
font-size: 12px;
}
```

✖ MISTAKE 2: Inconsistent Spacing

```
/* Wrong - random spacing */
padding: 10px 15px;
margin: 7px;
gap: 13px;

/* Correct - follows 4px grid /
padding: 12px 16px; / 3×4 and 4×4 /
margin: 8px; / 2×4 /
gap: 16px; / 4×4 */
```

✖ MISTAKE 3: Ignoring Dark Mode

```
/* Wrong - only light mode colors */
button { background-color: #E0E0E0; }

/* Correct - supports both modes */
@media (prefers-color-scheme: light) {
```

```
button { background-color: #E0E0E0; }
}

@media (prefers-color-scheme: dark) {
button { background-color: #424242; }
}
```

✖ MISTAKE 4: Over-Customizing Components

```
// Wrong - destroys Material Design
Card(
modifier = Modifier
.background(Color.Yellow)
.border(5.dp, Color.Magenta, RoundedCornerShape(45.dp))
) { }

// Correct - subtle customization
Card(
colors = CardDefaults.cardColors(
containerColor = MaterialTheme.colorScheme.primaryContainer
),
shape = RoundedCornerShape(12.dp),
elevation = CardDefaults.cardElevation(defaultElevation = 2.dp)
) { }
```

Figma & Design Tools

Rule 1: Material Design 3 Figma Kit

RULE: Use official Material Design 3 Figma components and libraries.

Setup:

1. Start with Material Design 3 UI Kit (available on Figma Community)
2. Enable shared libraries
3. Use components for consistency
4. Maintain component overrides carefully
5. Version components when changes made

Best Practices:

- Use main components, not instances directly
- Create local variants for customizations
- Document custom components clearly
- Keep components updated with releases

Rule 2: Design to Development Handoff

RULE: Structure Figma files for smooth developer handoff.

File Organization:

```
Design System/
├── Components/
│  ├── Button
│  ├── TextField
│  ├── Card
│  └── ...
├── Styles/
│  ├── Colors
│  ├── Typography
│  └── Spacing
├── Screens/
│  ├── Screen 1
│  ├── Screen 2
│  └── ...
└── Documentation/
└── Specs
```

Specifications:

- Use Inspect panel for measurements
- Export assets at correct scale (@1x, @2x, @3x)
- Document special interactions
- Provide color values in standard format (#RRGGBB, hex)

Version Information

- **Material Design Version**: 3 & 3 Expressive
- **Last Updated**: December 2025
- **Android Compose Material3**: Latest
- **Material Web**: Latest
- **Figma Kit**: Latest Material Design 3

References

- Material Design 3 Official Documentation
- Material Design 3 Foundations
- Material Theme Builder
- Compose Material 3 Documentation
- Material Web Documentation
- Material Design 3 Figma Kit
- WCAG 2.1 Guidelines

End of Document